

Static Timing Analysis

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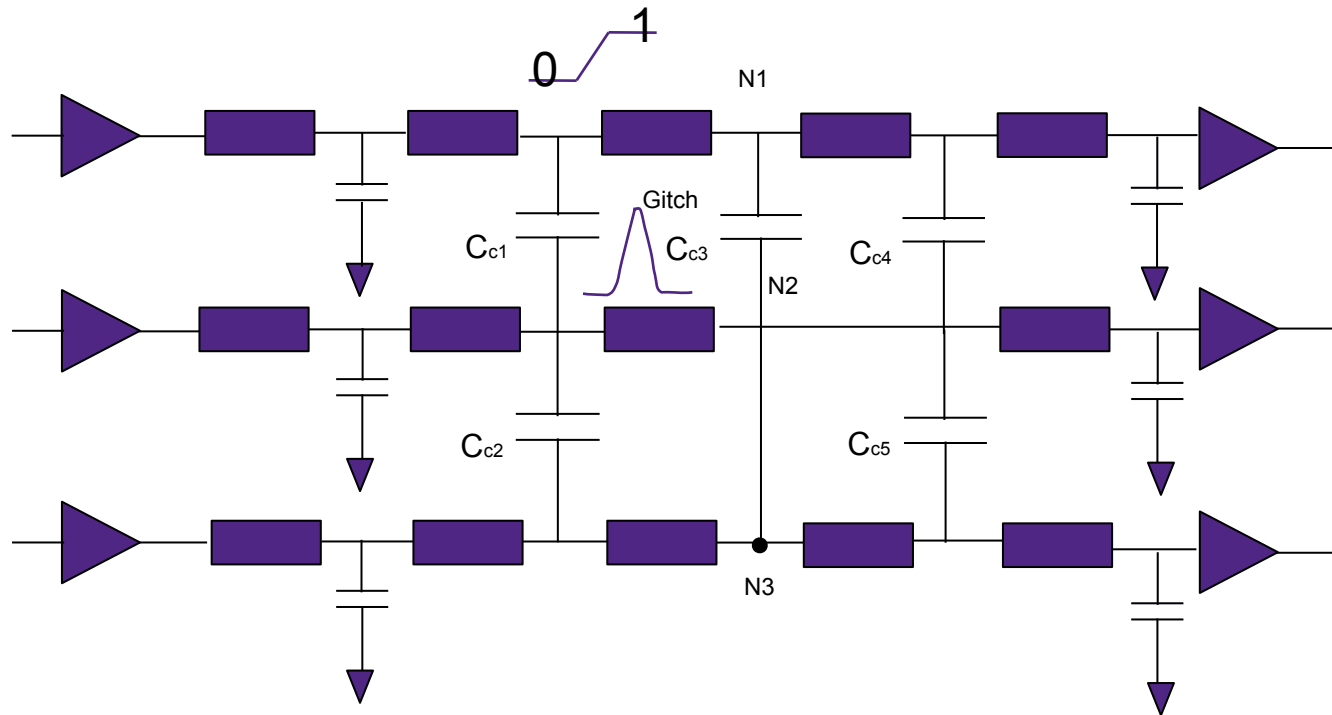
Course Overview

1. STA Concepts
 - 1 lecture
2. Delay Modeling
 - 1 lecture
3. Interconnect Parasitics
 - 1 lecture
4. Delay Calculation
 - 1 lecture
5. Configuring the STA Environment
 - 1 lecture
6. Timing Checks
 - 1 lecture
7. Crosstalk and Noise
 - 1 lecture



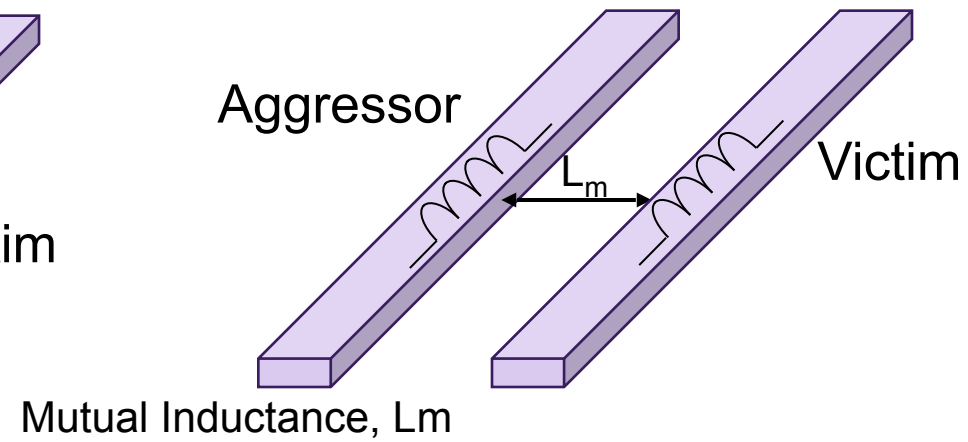
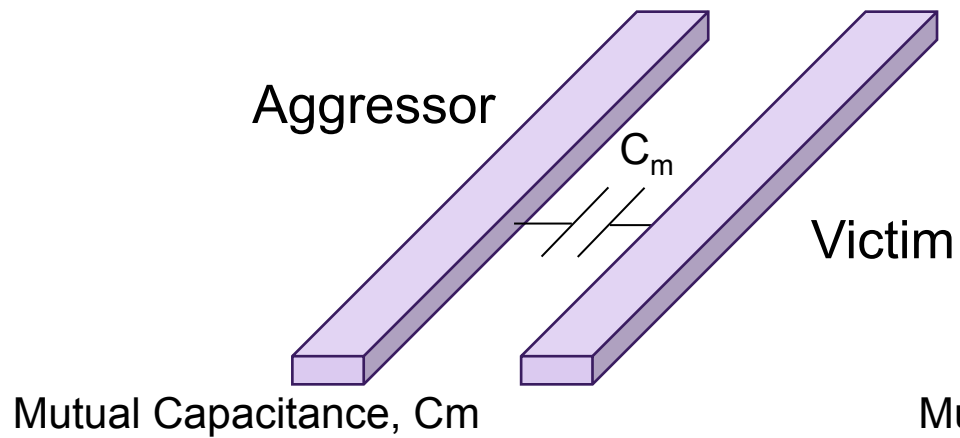
7. Crosstalk and Noise

Crosstalk

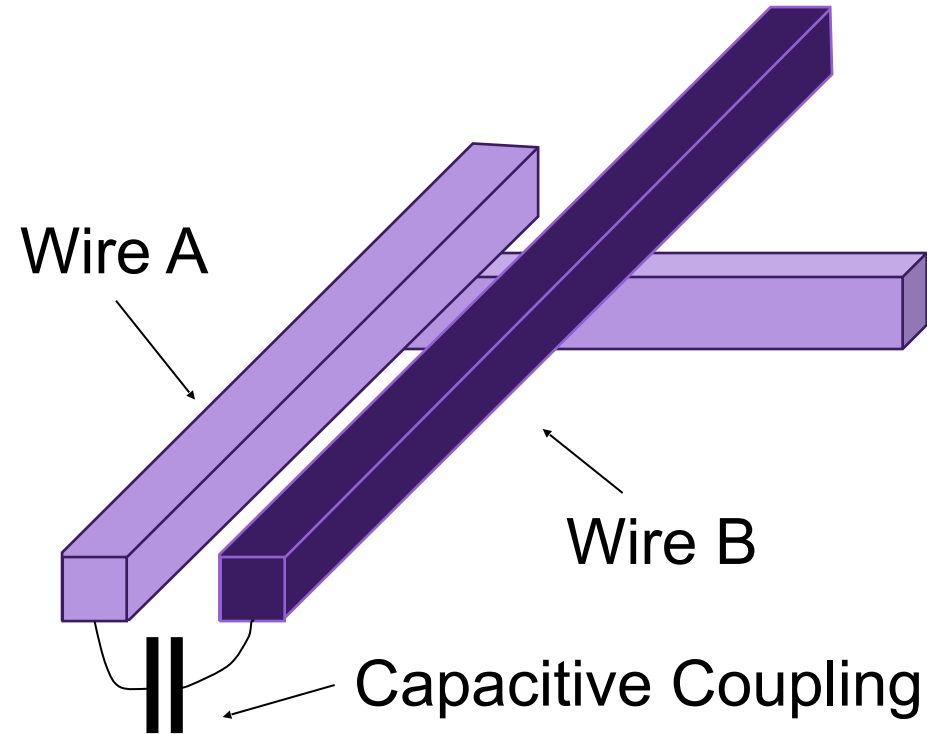


Crosstalk Definition

- Mechanisms that cause crosstalk
- Mutual capacitance
- Mutual inductance



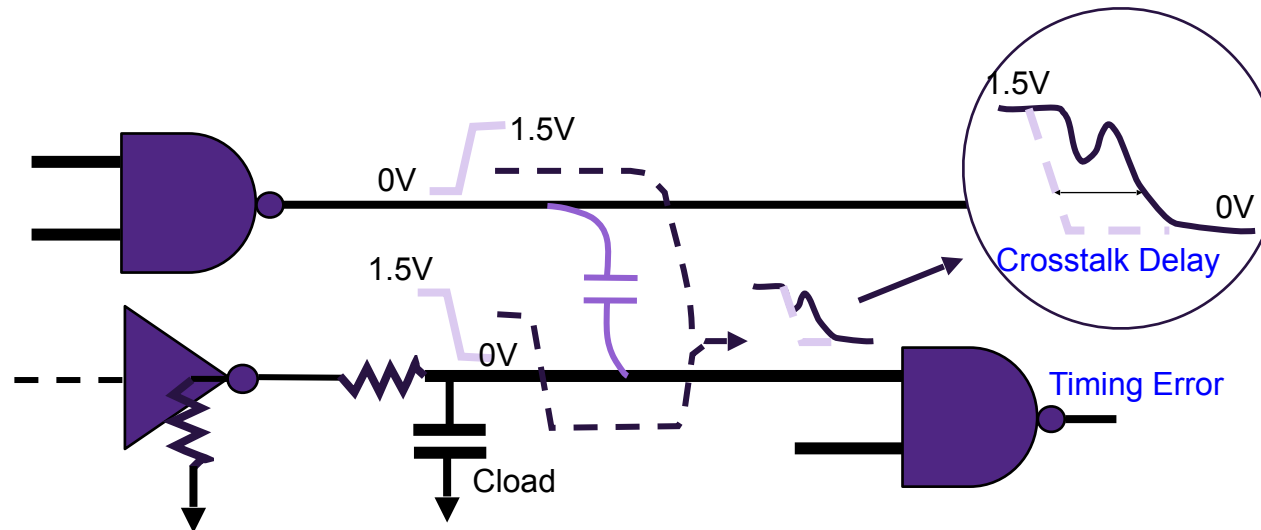
Capacitive Coupling



Noise Effects Types

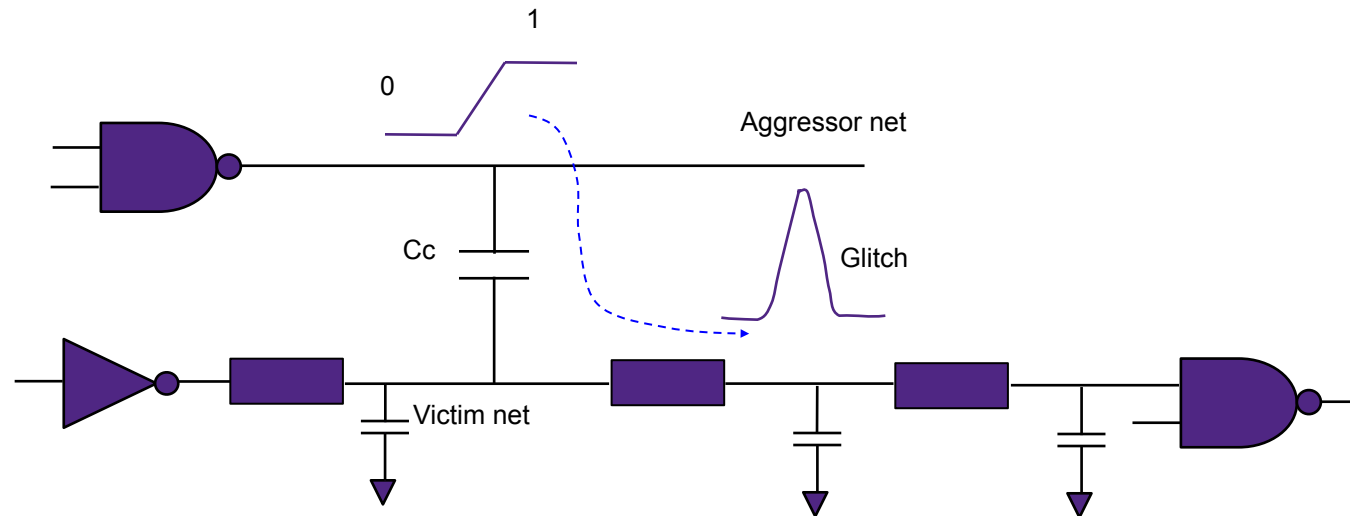
- Delay
 - Coupling of switching activity of the victim with the switching activity of the aggressors
- Glitch
 - Noise caused on a steady victim signal

Crosstalk Delay



Crosstalk Glitch

- A steady signal net can have a glitch due to charge transferred by the switching aggressors through the coupling capacitances

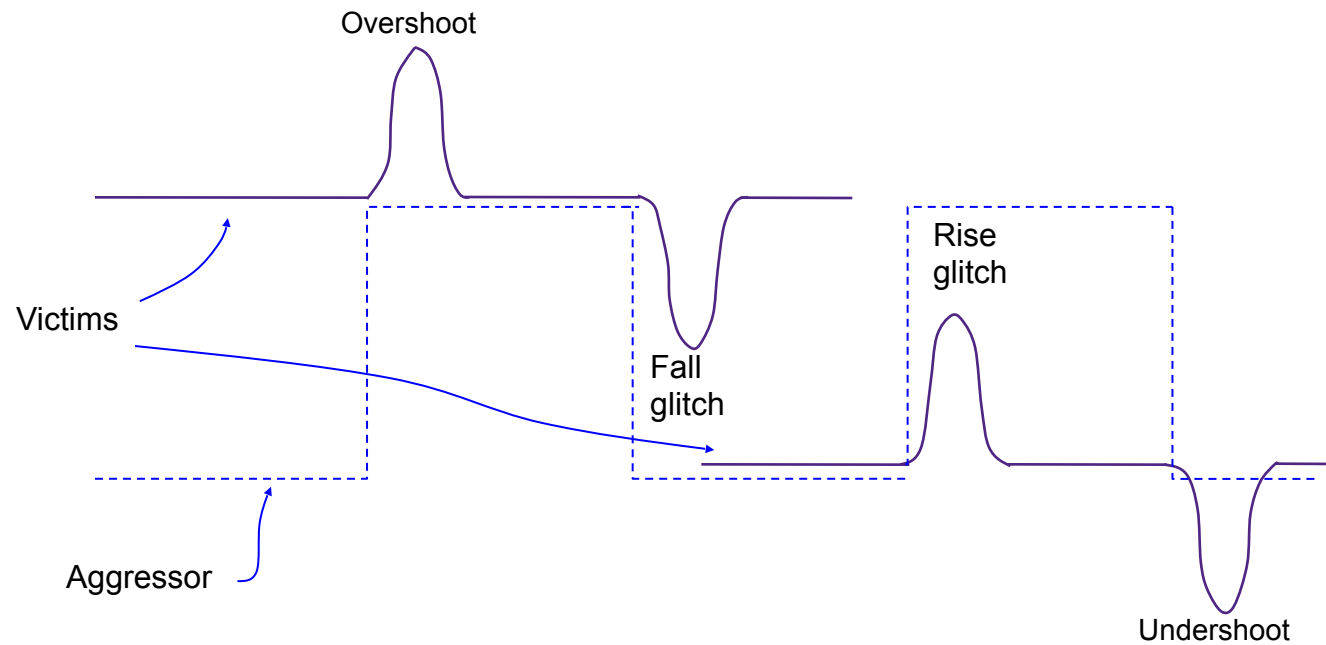


Factors Affecting Glitch

- Factors for large magnitude of glitch:
 - Large coupling capacitance
 - Fast slew time on the aggressor
 - Smaller victim net grounded capacitance
 - Smaller victim net driving strength

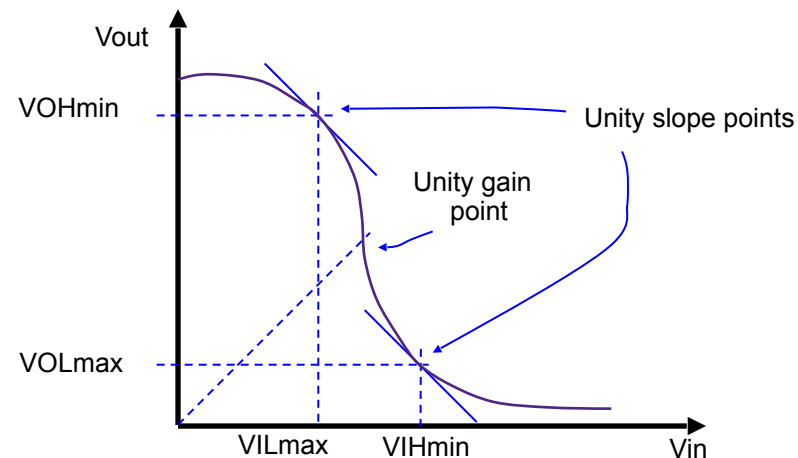
Types of Glitches

- Rise and fall glitches
- Overshoot and undershoot glitches



DC Noise Margins

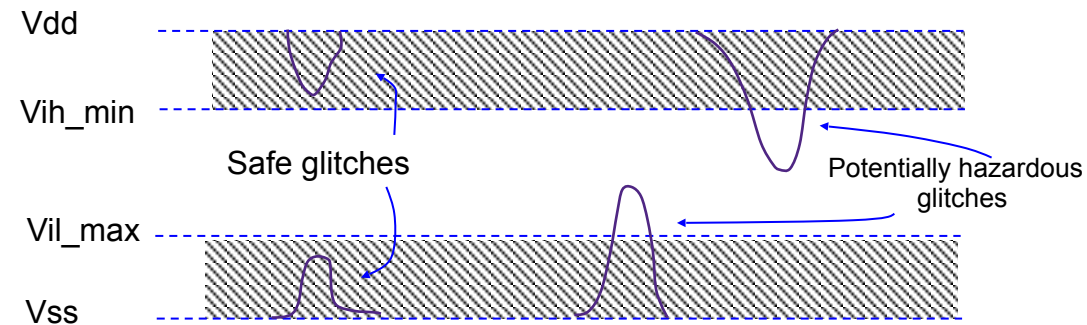
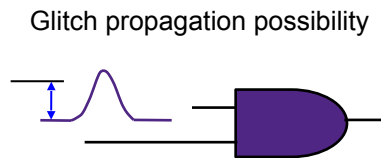
- DC noise limits on the input of a cell while ensuring proper logic functionality
- V_{IH} and V_{IL} are steady state noise limits
 - Whether a glitch will propagate through cell



DC characteristics of an inverter

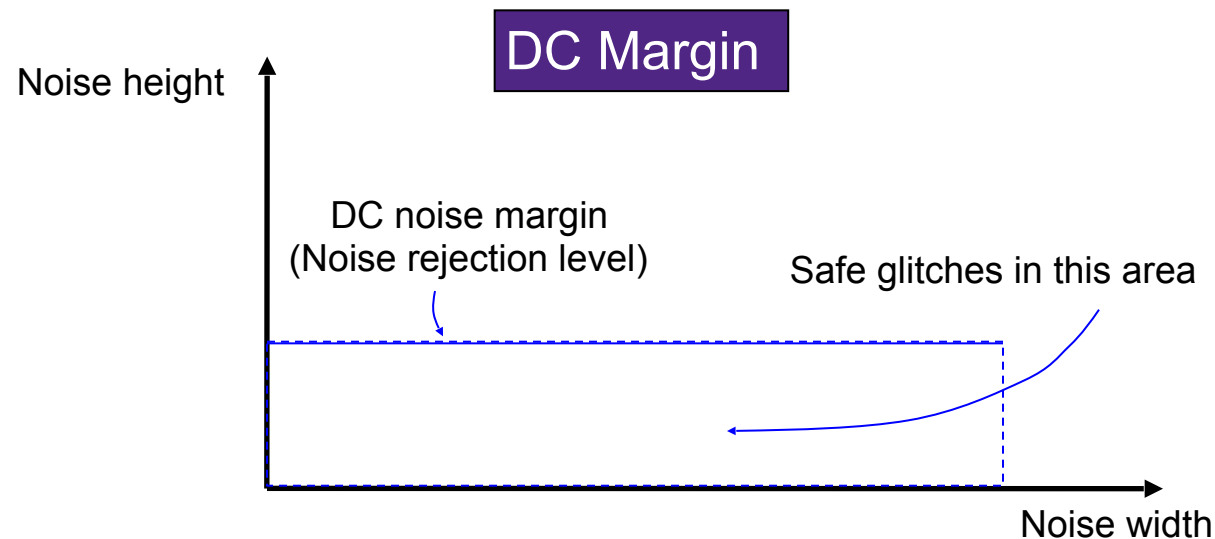
DC Margin Limits

- DC noise margin can be fixed to the same limit for all nets in the design
 - Glitch level is less than V_{ILmax} and greater than V_{IHmin}



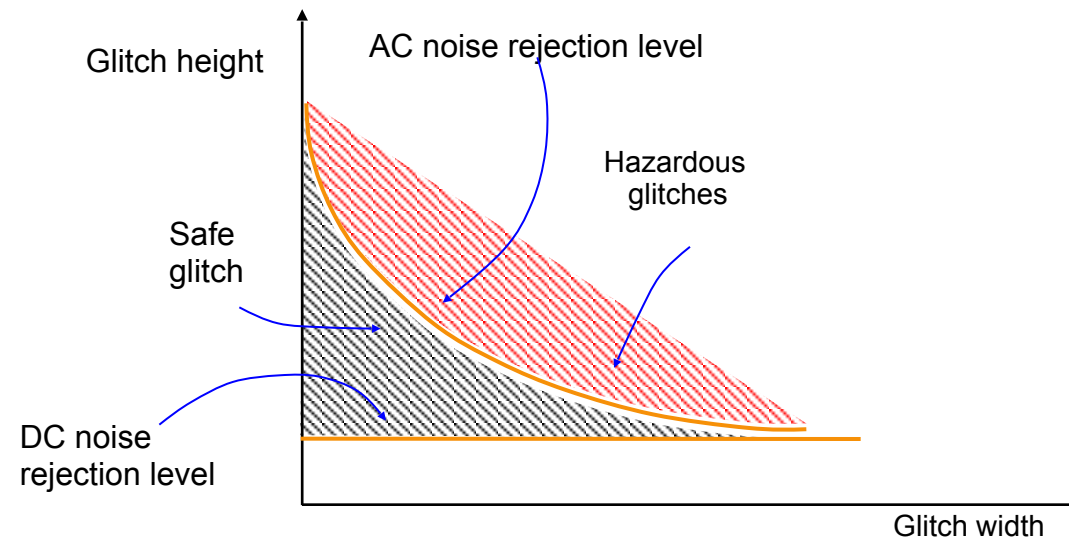
DC Margin Limits (2)

- Glitches magnitude is not enough to affect output of a cell
 - narrow width glitch → no effect
- DC margin checks only noise rejection level is used typically

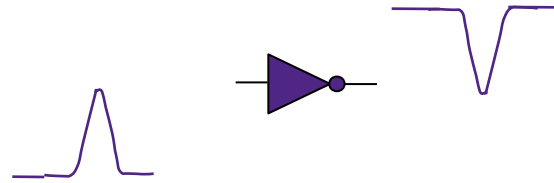


AC Noise Limits

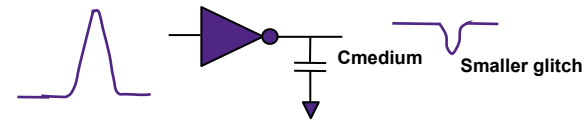
- Too narrow or too short glitches are safe
- AC noise rejection region can be specified



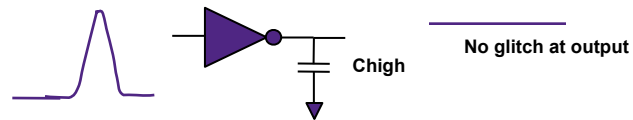
Basic Glitch Removal



No capacitive load



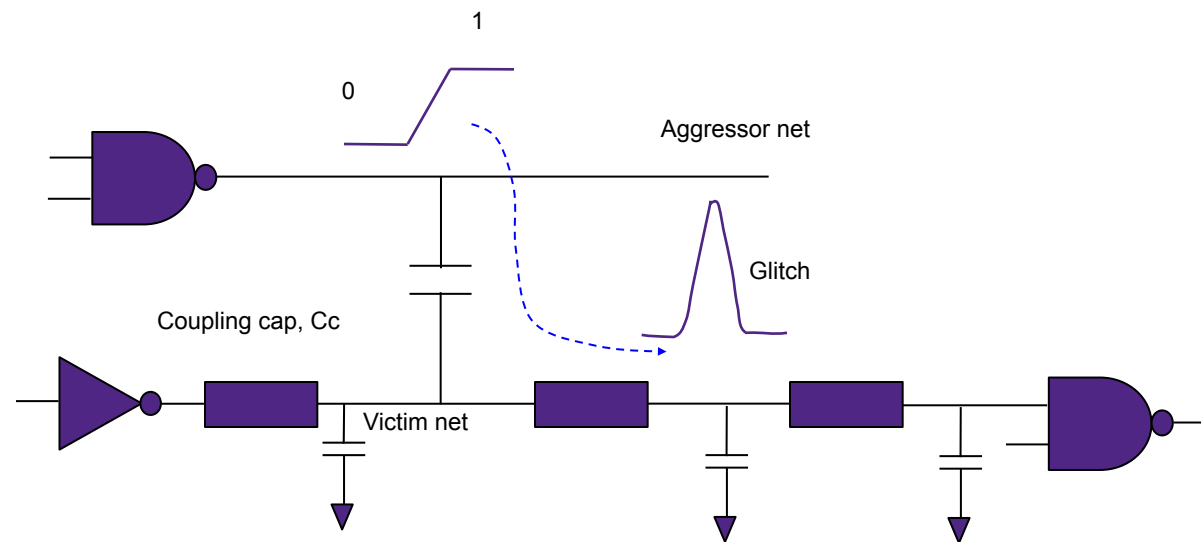
Medium capacitive load



Large capacitive load

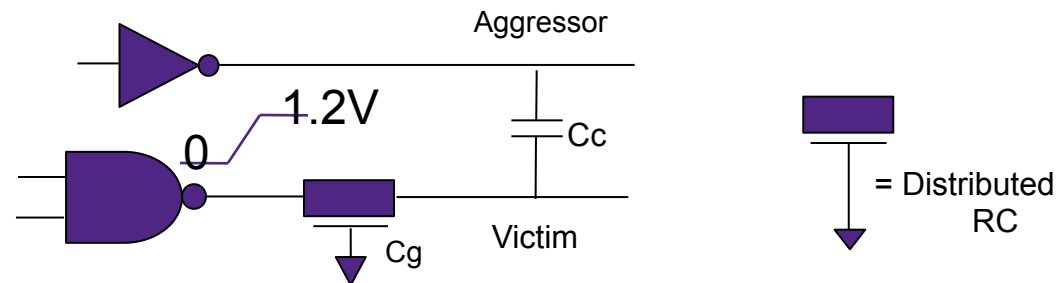
Multiple Aggressors

- Victim net may be capacitive-coupled to many nets
- The crosstalk coupling noise effect is accumulated due to multiple aggressors
- STA includes the timing correlation of the aggressor nets and determine multiple aggressors exist



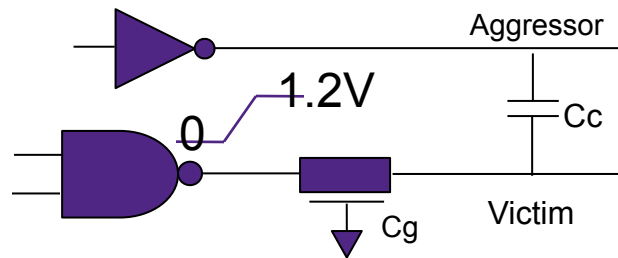
Crosstalk Delay Analysis

- Capacitance extraction for a net consists of different capacitances
 - Capacitance C_g to ground
 - Coupling capacitance C_c to a neighboring net



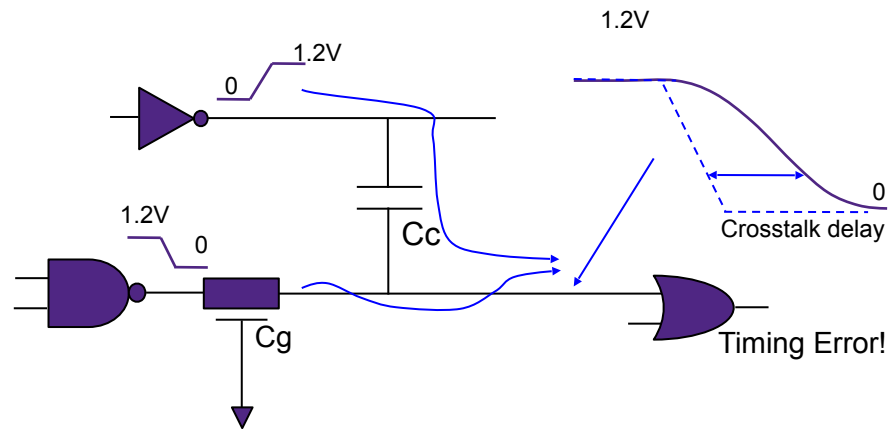
Crosstalk Scenarios

- Aggressor net stable
 - Victim net provides the charge for C_g and C_c to be charged to V_{dd}
 - Total charge = $(C_g + C_c) * V_{dd}$
- Aggressor switching in same direction
 - If aggressor slew is similar $\rightarrow$ total charge by the driving cell is $(C_g * V_{dd})$
 - If aggressor slew is faster $\rightarrow$ charge smaller than $(C_g * V_{dd})$
- Aggressor switching in opposite direction
 - C_c is charged from $-V_{dd}$ to V_{dd}
 - Charge on C_c changes by $(2 * C_c * V_{dd})$ before and after transitions



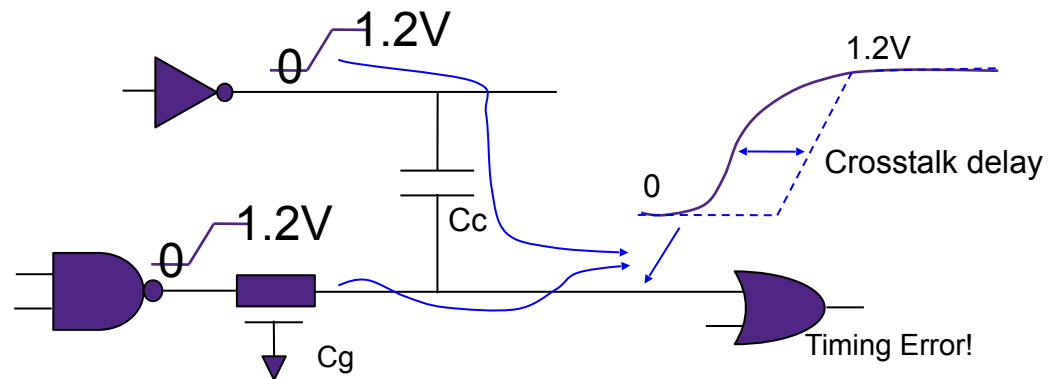
Positive Crosstalk

- Aggressor and victim are switching in opposite directions
 - Increases the amount of charge required from victim driver
 - Victim delay increases



Negative Crosstalk

- Aggressor and victim are switching in the same directions
 - Charge on C_c remains the same before and after transitions
 - Victim delay reduced

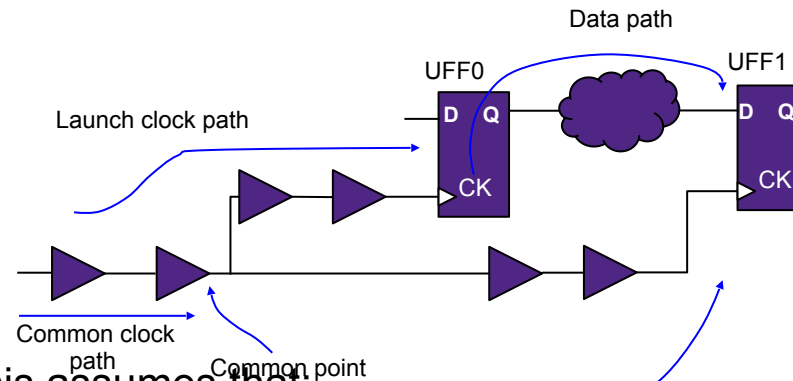


Crosstalk Timing Verification

- Positive rise delay
 - Rise edge moves forward in time
- Negative rise delay
 - Rise edge moves backward in time
- Positive fall delay
 - Fall edge moves forward in time
- Negative fall delay
 - Fall edge moves backward in time

Setup Analysis

- Consider the logic shown below where crosstalk can occur at various nets along the data path and along the clock paths.



- The setup (or max path) analysis assumes that:
 - Launch clock path sees positive crosstalk delay so that the data is launched late.
 - Data path sees positive crosstalk delay so that it takes longer for the data to reach the destination.
 - Capture clock path sees negative crosstalk delay so that the data is captured early.

Hold Analysis

- The worst condition for hold check occurs when both the launch clock path and the data path have negative crosstalk and the capture clock path has positive crosstalk. There is one important difference between the hold and setup analyses related to crosstalk on the common portion of the clock path.
- The worst-case hold (or min path) analysis for STA with crosstalk assumes:
 - Launch clock (not including the common path) sees negative crosstalk delay so that the data is launched early.
 - Data path sees negative crosstalk delay so that it reaches the destination early.
 - Capture clock (not including the common path) sees positive crosstalk delay so that the data is captured by the capture flipflop late.

Crosstalk Analysis for Complex ICs

- **Hierarchical Design and Analysis**
 - Checks that there be no coupling between signals inside the hierarchical block
 - Nets routed near boundary of the block should be shielded or avoided
- **Filtering of Coupling Capacitances**
 - Small value (below 1fF)
 - Coupling ratio (below 0.001)
 - Lumping small aggressors together (multiple aggressors with very small contributions)

Thank You